Fundamental Modules - 4

WEEK 4: ELECTRICAL SYSTEMS (AWARENESS LEVEL)

Days 19-24 | Detailed Hour-by-Hour Structure

Week Theme: *"Understanding the vehicle's nervous system - identification and basic testing only"*

DAY 19: ELECTRICAL FUNDAMENTALS + BASIC MULTIMETER USE

Learning Objectives:

- Understand basic electrical theory (Ohm's Law awareness)
- Identify circuit types and components
- Use multimeter for voltage testing only
- Understand fuses, relays, and protection devices

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Basic Electrical Theory

Electricity Basics:

- **Electricity = Flow of electrons through conductor**
- Like water flowing through pipe
- Needs complete circuit (path) to flow

Three Fundamental Quantities:

1. Voltage (V - Volts):

- Electrical "pressure"
- Pushes electrons through circuit
- Like water pressure in pipe
- **Automotive: 12V DC (Direct Current)**
- Battery provides voltage

2. Current (A - Amperes/Amps):

- Rate of electron flow
- Like volume of water flowing
- More current = more work done
- **Example:** Starter motor draws 150-300A
- Headlight bulb draws 5A
- LED light draws 0.5A

3. Resistance (Ω - Ohms):

- Opposition to current flow
- Like restriction in pipe
- Every component has resistance
- Wires have low resistance
- Bulbs, motors have higher resistance

Ohm's Law (Awareness Level):

- **V = I × R** (Voltage = Current × Resistance)
- **Understanding relationships:**
  - Higher voltage = more current (if resistance same)
  - Higher resistance = less current (if voltage same)
  - If current too high = wire overheats = fire risk

Example:

- Headlight bulb: 12V, 2.4Ω resistance
- Current = 12V ÷ 2.4Ω = 5A
- This is why we size wires and fuses correctly!

DC vs AC:

- **DC (Direct Current):** Flows one direction
  - All automotive 12V systems
  - Battery produces DC
- **AC (Alternating Current):** Reverses direction
  - Household electricity
  - Alternator produces AC, converted to DC

Conductors vs Insulators:

- **Conductors:** Allow current flow
  - Copper, aluminum (wires)

    - Low resistance
- **Insulators:** Block current flow
    - Rubber, plastic (wire insulation)
    - High resistance
    - Prevent short circuits

Hour 2 (10:00 AM - 11:00 AM): Automotive Circuits

Circuit Types:

1. Series Circuit:

- Components connected end-to-end
- Current flows through each component
- Same current through all components
- Voltage divides across components
- **Example:** Multiple lights on one switch
- **Problem:** One bulb fails = all fail

2. Parallel Circuit:

- Components connected side-by-side
- Each component has direct path to battery
- Same voltage across all components
- Current divides between branches
- **Example:** Each headlight independent
- **Advantage:** One fails, others still work

3. Series-Parallel (Combined):

- Mix of both types
- Most automotive circuits are this type
- Allows control and independence

Complete Circuit Requirements:

1. **Power source:** Battery (positive +)
2. **Load:** Device that uses electricity (bulb, motor)
3. **Path:** Wires connecting them
4. **Ground:** Return path to battery (negative -)
5. **Control:** Switch to turn on/off

Automotive Ground System:

- Negative battery terminal connected to chassis
- **Chassis = ground = negative return path**
- Saves wire (don't need negative wire to each component)
- Component bolted to chassis = grounded
- **Poor ground = component doesn't work properly**

Circuit Protection:

Fuses:

- **Purpose:** Protect wires from overheating/fire
- Thin wire inside melts if current too high
- Sacrificial component (designed to fail)
- **Types:**
  - **Blade/Spade fuse:** Most common, different colors
    - Mini (10-30A typical)
    - Standard ATO/ATC (5-30A)
    - Maxi (30-80A)
  - **Cartridge fuse:** Older vehicles, glass tube
  - **Fusible link:** Heavy wire link, engine bay
- **Ratings:** Amperage (5A, 10A, 15A, 20A, 30A, etc.)
- **Color codes:**
  - Tan = 5A
  - Brown = 7.5A
  - Red = 10A
  - Blue = 15A
  - Yellow = 20A
  - Clear/White = 25A
  - Green = 30A

Circuit Breakers:

- Resettable protection
- Used for high-draw circuits (power windows, wipers)
- Automatically reset when cooled
- Manual reset type exists

Relays:

- **Purpose:** Use small current to control large current
- Electromagnetic switch

- **Example:** Horn, headlights, fuel pump
- **Why needed:**
  - Dashboard switches can't handle high current
  - Small switch → relay → relay controls high current
- **Components:**
  - Coil (electromagnet)
  - Contacts (switch)
  - Usually 4 or 5 pins
- **Operation:**
  1. Small current energizes coil
  2. Coil creates magnetic field
  3. Magnetic field pulls contacts closed
  4. High current flows through contacts to load

Hour 3 (11:00 AM - 12:00 PM): Electrical Problems & Testing

Common Electrical Problems:

1. Open Circuit:

- Break in circuit path
- No current flows
- **Causes:**
  - Blown fuse
  - Broken wire
  - Corroded connector
  - Failed switch
- **Symptom:** Component doesn't work at all

2. Short Circuit:

- Unintended path to ground
- Excessive current flows
- **Causes:**
  - Wire insulation damaged
  - Wire touching chassis/metal
  - Internal component failure
- **Symptom:** Fuse blows immediately
- **Danger:** Can cause fire!

3. High Resistance:

- Partial blockage of current
- **Causes:**
  - Corrosion in connections
  - Loose connections
  - Damaged wire (partial break)
  - Poor ground connection
- **Symptom:** Component works poorly
  - Dim lights
  - Slow motor
  - Intermittent operation

4. Parasitic Drain:

- Something draws current when vehicle off
- **Causes:**
  - Interior light staying on
  - Module not sleeping
  - Short circuit
- **Symptom:** Dead battery after sitting

Electrical Testing Safety:

- **Never create shorts intentionally**
- **Disconnect battery when working on circuits**
- **Don't probe computer connectors randomly**
- **Use correct fuse rating (never higher!)**
- **Watch for sparks (indicates short or high current)**
- **Remove jewelry (conducts electricity)**

Basic Troubleshooting Logic:

- Component doesn't work:
  1. Check fuse first (easiest)
  2. Check power at component (battery voltage present?)
  3. Check ground (good connection to chassis?)
  4. Check switch operation
  5. Check component itself
- **Always start simple!**

Multimeter Basics (Introduction):

- **Purpose:** Measure voltage, current, resistance
- **For Fundamentals: Voltage testing only**

- **Settings:**
  - DC Voltage (V with straight line)
  - AC Voltage (V with wavy line) - not used in automotive 12V
  - Current (A) - not covering today
  - Resistance (Ω) - basic continuity only
- **Probes:**
  - Red = positive
  - Black = negative/ground
- **Safety:**
  - Never touch probe tips together (creates short)
  - Start on highest range if unsure
  - Auto-ranging meters easier

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Fuse & Relay Identification

Activity 1: Fuse Box Location (20 min)

- Locate all fuse boxes on vehicle:
  - **Interior fuse box:** Usually under dashboard or in side panel
  - **Engine bay fuse box:** Near battery or firewall
  - **Trunk fuse box:** Some vehicles (luxury/large)
- Identify fuse box components:
  - Fuse locations (numbered)
  - Fuse puller tool (usually clipped inside cover)
  - Spare fuses (stored in box)
  - Fuse diagram (inside cover or in manual)

Activity 2: Fuse Identification Practice (25 min)

- **Reading fuse diagram:**
  - Each fuse labeled with number
  - Function listed (Headlight, Radio, Fuel Pump, etc.)
  - Amperage rating shown (5A, 10A, etc.)
- **Removing and inspecting fuses:**
  - Use fuse puller (don't use fingers)
  - Hold fuse up to light
  - Look at thin wire inside:
    - **Intact = good** (wire continuous)

        - **Broken = blown** (wire melted, gap visible)
    - Check amperage rating on top
    - Check color matches rating
- **Fuse replacement practice:**
    - Match correct amperage
    - Match correct size (mini vs standard)
    - Push in firmly until seated
    - **Never use higher amperage fuse!** (fire risk)

Activity 3: Relay Location and Identification (15 min)

- Locate relays in fuse boxes:
    - Usually larger than fuses
    - Can be removed by pulling straight up
    - 4-pin or 5-pin most common
- Identify relay diagram:
    - Shows pin connections
    - Which circuits use relays
    - Common: Fuel pump, horn, headlights, A/C compressor
- **Listen to relay operation:**
    - Turn on component (horn, headlights)
    - Listen for click from relay
    - Clicking = relay working
    - No click = relay failed or no trigger signal
- **Relay swap test (if multiple identical):**
    - If suspect relay failure
    - Swap with identical relay
    - If problem moves = relay was bad
    - If problem stays = relay was good, problem elsewhere

Pass Criteria:

- ✅ Can locate all fuse boxes
- ✅ Can read fuse diagram
- ✅ Can remove and inspect fuses
- ✅ Can identify blown fuses
- ✅ Can replace fuses with correct rating
- ✅ Can locate and identify relays

Hour 2 (2:00 PM - 3:00 PM): Multimeter Voltage Testing Practice

Activity 1: Multimeter Familiarization (20 min)

- **Multimeter components:**
  - Display (shows reading)
  - Selection dial (choose function)
  - Probe ports:
    - COM (black probe - common/ground)
    - V/Ω (red probe - voltage/resistance)
    - A (current - not using today)
  - Red probe (positive)
  - Black probe (negative/ground)

Setting up for DC voltage:

1. Insert black probe into COM port
2. Insert red probe into V/Ω port
3. Turn dial to DC voltage (V with straight line or dash)
4. If manual range: Select 20V range (automotive is 12V)
5. Auto-range: Just select DC voltage

Safety check:

- Probes not touching each other
- Dial on correct setting
- Ready to measure

Activity 2: Battery Voltage Testing (25 min)

- **Battery voltage test procedure:**
  1. Set multimeter to DC voltage
  2. Touch black probe to negative (-) battery terminal
  3. Touch red probe to positive (+) battery terminal
  4. Read display

Interpreting battery voltage:

- **12.6V - 12.8V:** Fully charged (excellent)
- **12.4V - 12.6V:** Good charge (OK)
- **12.2V - 12.4V:** Low charge (recharge soon)
- **Below 12.2V:** Discharged (charge now)
- **Below 11.8V:** Deeply discharged (may be damaged)

Voltage test with engine running:

1. Start engine
2. Keep probes on battery terminals
3. Read voltage

Interpreting charging voltage:

- **13.5V - 14.5V:** Charging system working (good)
- **Above 14.8V:** Overcharging (voltage regulator problem)
- **Below 13.5V:** Not charging enough (alternator problem)
- **Same as key-off voltage:** Not charging at all

Practice on multiple vehicles:

- Students test different batteries
- Record voltage (engine off)
- Record voltage (engine running)
- Determine battery/charging system condition

Activity 3: Circuit Voltage Testing (15 min)

- **Testing voltage at a component:**
  - Example: Headlight bulb
  - Black probe to chassis ground (anywhere metal)
  - Red probe to component power wire
  - Should read ~12V when switched on
  - Less than 11V = voltage drop problem

Testing ground connection:

- Black probe to battery negative terminal
- Red probe to component ground point
- Should read < 0.5V when circuit operating
- Higher reading = poor ground

Finding voltage drops:

- High resistance connections show voltage drop
- Test along circuit to find where voltage is lost
- **NOTE:** Advanced diagnosis is for Diagnostics Track

Pass Criteria:

- ✅ Can set multimeter to DC voltage correctly
- ✅ Can connect probes correctly
- ✅ Can test battery voltage safely

- ✅ Can interpret battery voltage readings
- ✅ Can test charging system voltage
- ✅ Can test for voltage at components

Hour 3 (3:00 PM - 4:00 PM): Basic Continuity Testing

Activity 1: Understanding Continuity (15 min)

- **Continuity = complete electrical path**
- Used to check:
  - Fuses (good or blown)
  - Wires (intact or broken)
  - Grounds (connected or not)

Setting multimeter for continuity:

1. Turn dial to continuity symbol (usually diode symbol with sound waves)
2. Or use resistance (Ω) setting on lowest range
3. Touch probes together:
   - **Continuity mode:** Beeps (means connection)
   - **Resistance mode:** Reads 0 Ω or very low

Activity 2: Fuse Continuity Testing (20 min)

- **Testing fuses without removing:**
  - Touch probes to both metal tabs on fuse top
  - Should beep (or read 0Ω) if good
  - No beep (or OL/infinite) if blown
- **Testing removed fuses:**
  - Touch probes to each end of fuse
  - Beep = good
  - No beep = blown
  - Can see broken wire AND confirm with meter

Practice:

- Students test good and known-blown fuses
- Learn to recognize meter response
- Faster than visual inspection on some fuses

Activity 3: Ground Connection Testing (25 min)

- **Why ground testing matters:**
  - Poor ground = most common electrical problem
  - Component works poorly or not at all
  - Easy to test, easy to fix

Testing ground continuity:

1. Disconnect component from power
2. Touch one probe to component ground connection
3. Touch other probe to battery negative terminal
4. Should beep (or read < 1Ω)
5. No beep = open ground (bad connection)

Common ground problems:

- Rust/corrosion at ground point
- Loose bolt
- Broken wire
- Paint blocking connection

Finding and fixing poor grounds:

- Identify ground point (where wire bolts to chassis)
- Remove bolt
- Clean mating surfaces (wire brush, sandpaper)
- Remove paint if needed (metal-to-metal contact)
- Reinstall bolt tightly
- Retest continuity

Practice scenarios:

- Students locate ground points on vehicle
- Test continuity
- Identify poor grounds
- Clean and retest

NOTE: Students are TESTING only, not repairing wiring (that's Electrical Track)

Pass Criteria:

- ✅ Can set multimeter to continuity mode
- ✅ Can test fuses for continuity
- ✅ Can test ground connections
- ✅ Can identify poor ground connections

- ✅ Understands basic fixing process

DAY 19 ASSESSMENT:

Practical Skills Checklist:

- [ ] Locates all fuse boxes
- [ ] Reads fuse diagram correctly
- [ ] Removes and inspects fuses
- [ ] Identifies blown fuses visually
- [ ] Replaces fuses with correct rating
- [ ] Locates relays
- [ ] Sets multimeter to DC voltage
- [ ] Tests battery voltage correctly
- [ ] Interprets voltage readings
- [ ] Tests charging system voltage
- [ ] Sets multimeter to continuity
- [ ] Tests fuses for continuity
- [ ] Tests ground connections

Knowledge Check (Verbal):

- What is Ohm's Law relationship between voltage, current, resistance?
- What are the three types of circuits?
- What does a fuse do?
- What should battery voltage read when engine running?
- What is continuity?

DAY 20: BATTERY SYSTEMS + BATTERY TESTING

Learning Objectives:

- Understand battery construction and operation
- Identify battery specifications
- Perform battery testing (voltage and load test awareness)
- Practice safe battery handling and jump-starting

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Battery Fundamentals

Purpose of Battery:

- Store electrical energy chemically
- Provide power when engine off
- Provide surge current for starting (200-300A)
- Stabilize electrical system voltage
- Power electronics when demand exceeds alternator

Battery Construction:

- **Case:** Polypropylene plastic
  - Impact resistant
  - Chemical resistant
  - Vented (releases gases)
- **Cells:** 6 cells in series = 12V
  - Each cell = 2.1V
  - Separated by partitions
- **Plates:** Lead-based
  - **Positive plates:** Lead dioxide (brown)
  - **Negative plates:** Sponge lead (gray)
  - Multiple plates per cell (more plates = more capacity)
  - Separated by insulators (prevent contact)
- **Electrolyte:** Sulfuric acid + water solution
  - Specific gravity: 1.265 (fully charged)
  - Conducts electricity between plates
  - Participates in chemical reaction
- **Terminals:**
  - **Positive (+):** Larger diameter
  - **Negative (-):** Smaller diameter
  - Prevents incorrect connection

Chemical Reaction (Simplified):

- **Discharging (powering vehicle):**
  - Sulfuric acid + Lead plates → Lead sulfate + Water + Electricity
  - Acid becomes weaker (more water)
- **Charging (alternator running):**
  - Lead sulfate + Water + Electricity → Sulfuric acid + Lead plates

- Acid becomes stronger
- Reversible reaction

Battery Types:

1. Flooded (Wet Cell) - Traditional:

- Liquid electrolyte
- Removable caps (can add water)
- Requires maintenance
- Can spill if tipped
- Least expensive
- Most common

2. Maintenance-Free (Sealed):

- Flooded design but sealed
- Can't add water
- Less maintenance
- Still can spill if cracked
- Moderate cost

3. AGM (Absorbed Glass Mat):

- Electrolyte absorbed in glass mat
- Can't spill
- Better vibration resistance
- Higher current capacity
- More expensive
- Used in start-stop systems

4. Gel Cell:

- Electrolyte in gel form
- Can't spill
- Deep discharge tolerance
- More expensive
- Less common in automotive

Hour 2 (10:00 AM - 11:00 AM): Battery Specifications & Selection

Battery Ratings:

1. Cold Cranking Amps (CCA):

- Most important rating
- Current battery can deliver for 30 seconds at -18°C (0°F)
- While maintaining 7.2V minimum
- **Higher = better starting power**
- Typical: 400-800 CCA
- Must meet or exceed vehicle requirement

2. Reserve Capacity (RC):

- Minutes battery can deliver 25A at 27°C (80°F)
- While maintaining 10.5V minimum
- **How long vehicle runs if alternator fails**
- Typical: 60-120 minutes
- Higher = more emergency capacity

3. Amp-Hour (Ah) Rating:

- Total energy storage capacity
- Example: 60Ah = 60A for 1 hour, or 30A for 2 hours, etc.
- Used for calculating accessory loads
- Not always specified

Battery Group Size:

- Physical dimensions
- Terminal location
- **Common sizes:**
  - Group 24, 35, 51, 65, 75, etc.
  - Must fit vehicle battery tray
  - Must have terminals in correct position

Reading Battery Label:

- Manufacturer name
- CCA rating (most prominent)
- RC rating
- Group size
- Date code (manufacturing date)
- Voltage (12V)
- Type (Flooded, AGM, etc.)
- Warranty period

Battery Selection Criteria:

- **Must match or exceed:**
  - CCA requirement (never lower)
  - Physical size (group size)
  - Terminal type (top-post, side-post)
- **Consider:**
  - Warranty length
  - Brand reputation
  - AGM if start-stop vehicle
  - Budget

Battery Life Factors:

- **Normal life:** 3-5 years
- **Shortened by:**
  - High temperatures (hot climates)
  - Vibration (loose hold-down)
  - Deep discharges (leaving lights on)
  - Overcharging (bad voltage regulator)
  - Undercharging (short trips, bad alternator)
  - Sulfation (sitting discharged)

Hour 3 (11:00 AM - 12:00 PM): Battery Testing & Diagnosis

Visual Inspection:

- **Case condition:**
  - Cracks, leaks
  - Bulging (overcharged or frozen)
  - Clean and dry exterior
- **Terminals:**
  - Corrosion (white/green/blue powder)
  - Tight connection
  - Damaged/broken
- **Hold-down:**
  - Secure (prevents vibration damage)
  - Not over-tightened (can crack case)
- **Electrolyte level (if accessible):**
  - Should cover plates

  - Add distilled water if low (not tap water!)
  - If maintenance-free, can't check

Voltage Testing (Already Practiced):

- **Surface charge:** If just driven, let sit 15 min
- **Voltage reading interpretation:**
  - 12.6V+ = 100% charged
  - 12.4V = 75% charged
  - 12.2V = 50% charged
  - 12.0V = 25% charged
  - <11.8V = discharged
- **If low:** Charge and retest

Load Testing (Awareness Level):

- **Purpose:** Test battery under load (simulates cranking)
- **Equipment:** Carbon pile load tester
- **Procedure (observation only for fundamentals):**
  1. Battery must be fully charged (12.6V+)
  2. Apply load equal to 1/2 CCA for 15 seconds
  3. Voltage should stay above 9.6V
  4. Below 9.6V = weak battery, replace
- **Why it matters:**
  - Battery can show good voltage but fail under load
  - Load test reveals capacity
  - Most accurate test

Conductance Testing (Modern Method):

- Electronic tester
- Sends signal through battery
- Measures internal resistance
- Faster than load test
- Provides CCA reading
- Print-out with pass/fail

Hydrometer Testing (Traditional):

- For flooded batteries only
- Measures electrolyte specific gravity
- Higher SG = more charge
- Can identify dead cells (one cell reads much lower)

- Not possible on sealed/AGM batteries

Battery Failure Symptoms:

- **Slow cranking:** Tired battery
- **Clicking when starting:** Very weak battery
- **No crank, lights dim:** Dead battery
- **No crank, lights bright:** Starter problem (not battery)
- **Voltage drops rapidly:** Bad cell
- **Won't hold charge:** Sulfated or dead cell

When to Replace Battery:

- Failed load test
- Age > 5 years and showing symptoms
- Cracked or damaged case
- Bulging case
- Heavy corrosion that can't be cleaned
- Repeatedly going dead

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Battery Inspection & Testing

Activity 1: Battery Information Recording (20 min)

- Students locate and record from battery label:
  - Manufacturer
  - CCA rating
  - RC rating (if listed)
  - Group size
  - Type (flooded, AGM, etc.)
  - Date code (decode manufacturing date)
  - Warranty remaining?
- Compare to vehicle requirements:
  - Check owner's manual specification
  - Is battery adequate?
  - Original or replacement?

Activity 2: Visual Battery Inspection (20 min)

- **Systematic inspection:**

Case inspection:

- Cracks or leaks
- Bulging sides
- Cleanliness
- Battery tray condition (rust, corrosion)

Terminal inspection:

- Corrosion amount and location
- Loose connections (try to move cables)
- Damaged cables or clamps
- Terminal condition

Hold-down inspection:

- Present and secure
- Not over-tightened
- Bracket not broken

Electrolyte level (if flooded):

- Remove cell caps if accessible
- Check level (should cover plates)
- All cells same level?
- Color (should be clear, not brown)
- Create battery inspection report
- Rate: Good / Service Needed / Replace

Activity 3: Battery Voltage Testing Review (20 min)

- Review voltage testing from Day 19
- Students test multiple batteries:
  - Engine off voltage (state of charge)
  - Engine running voltage (charging system)
  - Record readings
  - Interpret condition
- Practice explaining results to "customer":
  - "Your battery is showing 12.1V, which indicates it's about 40% charged. This suggests either the battery is failing to hold a charge, or your charging system isn't working properly. We should test the charging system next."

Pass Criteria:

- ✅ Can locate battery specifications on label
- ✅ Can decode battery information
- ✅ Performs thorough visual inspection
- ✅ Can test battery voltage correctly
- ✅ Interprets voltage readings accurately
- ✅ Creates professional inspection report

Hour 2 (2:00 PM - 3:00 PM): Battery Maintenance & Cleaning

Activity 1: Terminal Corrosion Cleaning (30 min)

Why terminals corrode:

- Sulfuric acid fumes escape
- React with metal terminals
- Creates powdery deposits
- Increases resistance (poor connection)

Cleaning procedure:

1. **Safety:**
   - Wear safety glasses
   - Gloves recommended
   - Work in ventilated area
2. **Disconnect battery (review from Week 1):**
   - Negative terminal FIRST
   - Then positive terminal
   - Keep cables away from terminals
3. **Clean terminals:**
   - Use wire brush or terminal cleaner tool
   - Scrub both terminals (battery side)
   - Scrub both cable clamps (inside)
   - Clean until shiny metal visible
   - If very corroded: Baking soda + water solution
     - Neutralizes acid
     - Bubbles = neutralizing reaction
     - Rinse with water after
4. **Clean battery top:**
   - Wipe with damp cloth
   - Remove any spilled acid or corrosion

    - Dry thoroughly
5. **Reconnect battery:**
    - Positive terminal FIRST
    - Then negative terminal
    - Tighten securely (don't over-tighten)
6. **Apply protection:**
    - Terminal grease or anti-corrosion spray
    - Prevents future corrosion
    - Thin coating on terminals and clamps

Students practice on batteries with corrosion:

- Each student cleans a battery
- Instructor supervises
- Proper technique emphasized
- Safety enforced

Activity 2: Battery Hold-Down Check (15 min)

- Inspect hold-down bracket:
    - Type (top-mount, side-mount, strap)
    - Secure but not over-tight
    - Bracket not damaged
- Check battery movement:
    - Try to move battery by hand
    - Should have no movement
    - Movement = vibration damage risk
- Tighten if loose:
    - Use correct tool
    - Secure but don't crack case

Activity 3: Water Addition (Flooded Batteries) (15 min)

- **NOTE:** Only for flooded batteries with removable caps
- **When needed:** Electrolyte level below plates
- **Procedure:**
    1. Remove cell caps
    2. Check level in each cell
    3. Add **distilled water** only (not tap water!)
        - Tap water has minerals (damage battery)
    4. Fill to indicator or just above plates
    5. Don't overfill

6. Reinstall caps
7. Charge battery after adding water

- **Modern batteries:** Most are sealed, can't add water

Pass Criteria:

- ✅ Can clean battery terminals safely
- ✅ Disconnects/reconnects battery correctly
- ✅ Uses proper cleaning materials
- ✅ Checks and secures hold-down
- ✅ Knows when and how to add water
- ✅ Applies corrosion protection

Hour 3 (3:00 PM - 4:00 PM): Jump-Starting & Battery Safety

Activity 1: Jump-Starting Procedure (25 min)

When to jump-start:

- Battery too weak to crank engine
- Interior light left on overnight
- Battery near end of life
- Cold weather (reduces capacity)

Equipment needed:

- Jumper cables (heavy gauge, 4-6 gauge)
- Good battery (another vehicle or jump pack)
- Safety glasses

CRITICAL SAFETY:

- Batteries produce hydrogen gas (explosive!)
- Any spark can cause explosion
- Follow procedure exactly
- Never lean over battery during jump

Jump-starting procedure (CRITICAL SEQUENCE):

1. **Position vehicles:**
   - Close enough for cables to reach
   - NOT touching (prevent short)
   - Both in Park/Neutral, parking brake on

- Both engines OFF initially

2. **Connect cables in ORDER:**
   - **① RED clamp → DEAD battery POSITIVE (+)**
   - **② RED clamp → GOOD battery POSITIVE (+)**
   - **③ BLACK clamp → GOOD battery NEGATIVE (-)**
   - **④ BLACK clamp → DEAD vehicle ENGINE BLOCK (not battery!)**
     - Use bare metal, away from battery
     - This is the GROUND connection
     - Any spark occurs here (safer)
3. **Start good vehicle:**
   - Run for 2-3 minutes
   - Allows some charge to transfer
4. **Start dead vehicle:**
   - Crank for 10 seconds max
   - If doesn't start, wait 1 minute and retry
   - If still doesn't start after 3 tries, problem isn't just battery
5. **Disconnect cables in REVERSE ORDER:**
   - **④ BLACK from engine block (dead vehicle)**
   - **③ BLACK from good battery negative**
   - **② RED from good battery positive**
   - **① RED from dead battery positive**
6. **Keep dead vehicle running:**
   - Drive for 20-30 minutes to recharge
   - Don't turn off immediately

Common mistakes (dangerous!):

- Connecting negative to dead battery (spark near battery)
- Cables on backwards (can damage electronics)
- Cables touching each other while connected
- Leaning over battery during connection

Students practice (engine OFF, just cable connection):

- Instructor sets up scenario
- Students connect cables correctly
- Verbalize each step
- Explain why sequence matters

Activity 2: Jump Pack Usage (15 min)

- **Portable jump starters (jump packs):**

    - Self-contained battery and cables
    - No second vehicle needed
    - Same procedure:
        1. RED to dead positive
        2. BLACK to engine block (ground)
        3. Turn on jump pack
        4. Start vehicle
        5. Disconnect in reverse
- **Advantages:**
    - Convenient, portable
    - No second vehicle needed
    - Safer (reverse polarity protection on most)
- Students practice with jump pack (if available)

Activity 3: Battery Safety & Disposal (20 min)

Battery safety rules:

- **Explosive gas:** No sparks, flames, smoking near battery
- **Acid burns:** If spilled on skin, flush with water immediately
- **Eye protection:** Always wear safety glasses
- **Ventilation:** Work in open area
- **Heavy:** Use proper lifting technique
- **Secure:** Always use hold-down
- **Correct polarity:** Double-check connections

What NOT to do:

- Never test by shorting terminals (spark!)
- Never charge frozen battery (can explode)
- Never overcharge (gas buildup)
- Never use near open flame
- Never add acid (add distilled water only)

Battery disposal:

- **NEVER throw in trash** (lead and acid are toxic)
- **Recycle at:**
    - Auto parts stores (usually take for free)
    - Battery retailers (usually pay small amount)
    - Recycling centers
- **Core charge:** Pay deposit when buying new, get back when returning old

Understanding battery recycling:

- 98% recyclable
- Lead reclaimed for new batteries
- Plastic recycled
- Acid neutralized and recycled
- Environmental responsibility

Pass Criteria:

- ✅ Can perform jump-start procedure correctly
- ✅ Connects cables in correct sequence
- ✅ Explains why sequence matters
- ✅ Can use jump pack safely
- ✅ Understands battery safety rules
- ✅ Knows proper disposal methods

DAY 20 ASSESSMENT:

Practical Skills Checklist:

- [ ] Reads battery specifications
- [ ] Performs visual battery inspection
- [ ] Tests battery voltage correctly
- [ ] Interprets voltage readings
- [ ] Cleans battery terminals safely
- [ ] Disconnects/reconnects battery correctly
- [ ] Checks battery hold-down
- [ ] Performs jump-start procedure correctly
- [ ] Uses jump pack safely
- [ ] Follows battery safety rules

Knowledge Check (Verbal):

- What does CCA stand for and why is it important?
- What voltage indicates a fully charged battery?
- Why do we connect negative jumper cable to engine block instead of battery?
- What should you do with a dead battery?
- How often should batteries typically be replaced?

DAY 21: STARTING & CHARGING SYSTEMS + BASIC TESTING

Learning Objectives:

- Understand starter motor operation
- Understand alternator operation and voltage regulation
- Identify starting and charging components
- Perform basic charging system tests

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Starting System

Purpose of Starting System:

- Convert electrical energy (battery) to mechanical energy
- Rotate engine for initial combustion
- Must spin engine 150-250 RPM minimum for start

Starter Motor Construction:

- **Electric motor:** DC motor design
- **Components:**
  - **Armature:** Rotating part with windings
  - **Field coils:** Create magnetic field (or permanent magnets)
  - **Commutator:** Switches current direction in armature
  - **Brushes:** Conduct current to commutator
  - **Drive gear (pinion):** Small gear that engages flywheel
  - **Solenoid:** Electromagnetic switch
    - Pushes pinion into engagement
    - Closes high-current contacts

Starting System Components:

1. Battery:

- Provides 200-300A surge current
- Must be fully charged

2. Ignition Switch:

- Driver control

- Sends 12V signal to starter relay/solenoid

3. Starter Relay (some vehicles):

- Allows small ignition switch to control high current
- Protects ignition switch from wear

4. Neutral Safety Switch (automatic) / Clutch Switch (manual):

- Prevents starting unless in Park/Neutral or clutch depressed
- Safety feature

5. Heavy cables:

- Positive cable from battery to starter (thick, 4-gauge or larger)
- Ground from battery to engine/chassis

6. Starter motor: Main component

Starting System Operation:

1. Turn key to START position
2. Ignition switch sends 12V to starter solenoid
3. Solenoid energizes:
    - Plunger pushes pinion gear forward
    - Pinion engages with flywheel ring gear
    - Internal contacts close (high current path)
4. Current flows from battery through starter
5. Starter motor spins (200A+ current draw)
6. Pinion spins flywheel
7. Flywheel spins crankshaft
8. Engine starts
9. Release key
10. Solenoid de-energizes
11. Pinion retracts (overrunning clutch prevents damage)

Common Starting Problems:

Click, no crank:

- Weak battery (most common)
- Corroded connections
- Faulty starter solenoid

No click, no crank:

- Dead battery
- Ignition switch failure
- Neutral safety switch
- Broken wire

Slow cranking:

- Weak battery
- Corroded connections
- Worn starter motor
- Engine mechanical problem (seized, thick oil)

Grinding noise:

- Worn starter drive gear
- Worn flywheel teeth
- Improper engagement

Starter stays engaged:

- Stuck solenoid
- Seized starter drive
- **Turn off immediately!** (can damage)

Hour 2 (10:00 AM - 11:00 AM): Charging System

Purpose of Charging System:

- Recharge battery after starting
- Power all electrical systems when engine running
- Maintain voltage at 13.5-14.5V

Alternator Operation:

- **Alternator = AC generator**
- Produces AC (alternating current)
- Diodes convert AC to DC for vehicle use

Alternator Construction:

- **Rotor:** Rotating electromagnet inside
    - Spun by engine belt
    - Field current creates magnetic field

- **Stator:** Stationary windings outside rotor
  - AC voltage induced in windings
  - Three-phase AC output
- **Diode rectifier:** Converts AC to DC
  - Six diodes (diode trio/bridge)
  - Block reverse current
- **Voltage regulator:** Maintains constant voltage
  - Internal (modern) or external (older)
  - Varies field current to control output
  - Target: 13.5-14.5V
- **Pulley:** Belt-driven from crankshaft
- **Cooling fan:** Cools alternator internally

Charging System Components:

1. Battery: Stores energy, smooths voltage

2. Alternator: Generates electricity

3. Drive belt (serpentine): Spins alternator

- Must be tight and in good condition
- Slipping belt = poor charging

4. Wiring:

- Heavy output wire (battery +)
- Excitation wire (field current)
- Sense wire (reads battery voltage)
- Ground connection

5. Battery warning light / gauge:

- Dashboard indicator
- Lights if charging problem
- Usually symbol of battery

Charging System Operation:

1. Engine runs, belt spins alternator
2. Initial field current from battery energizes rotor
3. Spinning rotor creates changing magnetic field
4. Changing field induces AC in stator windings
5. Diodes rectify AC to DC

6. DC output sent to battery and vehicle electrical
7. Voltage regulator monitors battery voltage
8. If voltage low: Increases field current (more output)
9. If voltage high: Decreases field current (less output)
10. Maintains steady 13.5-14.5V

Alternator Output:

- Rated in amperes (A)
- Common: 60A, 80A, 100A, 120A, 150A
- Must meet vehicle's electrical demand
- Higher demand vehicles need higher output

Voltage Regulation:

- **Too low (<13.5V):** Battery won't charge
- **Correct (13.5-14.5V):** Optimal charging
- **Too high (>14.8V):** Overcharging
  - Boils battery
  - Damages electronics
  - Burns out bulbs

Hour 3 (11:00 AM - 12:00 PM): Charging System Problems

Common Charging Problems:

1. No Charge:

- **Symptoms:**
  - Battery light stays on
  - Voltage doesn't rise when engine starts
  - Battery goes dead
- **Causes:**
  - Broken belt (alternator not spinning)
  - Failed alternator
  - Blown fusible link
  - Bad voltage regulator
  - Wiring problems

2. Undercharging:

- **Symptoms:**

    - Voltage 12.5-13.0V (too low)
    - Battery slowly drains
    - Dim lights at idle
- **Causes:**
    - Slipping belt
    - Failing alternator
    - Worn brushes
    - Bad diode (one phase out)
    - High resistance in wiring

3. Overcharging:

- **Symptoms:**
    - Voltage >14.8V
    - Battery boiling
    - Bulbs burning out
    - Strong electrical smell
- **Causes:**
    - Failed voltage regulator
    - Bad ground connection
    - Regulator sense wire disconnected

4. Noisy Alternator:

- **Grinding:** Worn bearings (common failure)
- **Whining:** Bad diode (electrical noise through radio)
- **Squealing:** Slipping belt

Diagnosing Charging Problems:

Visual checks:

- Belt condition and tension
- Wiring connections tight
- No burning smell
- Alternator mounting secure

Voltage test (already learned):

- Engine off: 12.6V
- Engine running: 13.5-14.5V
- Too low or too high = problem

Belt test:

- Inspect for cracks, fraying, glazing
- Check tension (1 cm deflection typical)
- Check for oil contamination (causes slipping)

Charging output test (awareness level):

- Test voltage at battery with engine running
- Turn on high loads (headlights, blower, rear defrost)
- Voltage should stay above 13.0V
- Drops below = alternator can't keep up

Alternator amperage test (load test):

- Requires special tester
- Measures actual amp output
- Should meet rating (within 10%)
- NOT performed in fundamentals (Electrical Track)

When to Replace Alternator:

- No output (failed)
- Overcharging (regulator failed)
- Worn bearings (noise)
- Insufficient output even at high RPM
- Age + declining performance

When to Replace Belt:

- Cracks, fraying, chunks missing
- Glazed (shiny surface from slipping)
- Stretching (can't tension properly)
- Oil-soaked
- Age (preventive, 60-100k km)

Jump-Starting Effects:

- Improper jump-starting can damage alternator
- Voltage spikes from disconnecting cables
- Always careful with polarity

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Starting & Charging System Identification

Activity 1: Starting System Component Location (30 min)

- Identify all starting system components:

Battery: (already familiar)

- Location, cable connections

Starter motor:

- Location: Usually bottom of engine, near flywheel
- Size: Cylinder about 15-20cm long
- Solenoid: Mounted on starter body
    - Has 2-3 terminals (battery +, ignition +, ground)

Heavy positive cable:

- From battery + to starter solenoid
- Very thick (4-6 gauge)
- Often red (but may be black)

Ignition signal wire:

- Smaller wire to solenoid
- Activates starter when key turned

Starter relay (if present):

- In fuse box or on firewall
- Check diagram

Neutral safety switch:

- On transmission (automatic)
- Or clutch pedal (manual)
- May not be visible
- Draw starting system diagram
- Trace current flow from battery to starter

Activity 2: Charging System Component Location (30 min)

- Identify all charging system components:

Alternator:

- Location: Front of engine, belt-driven
- Pulley visible

- Cooling vents visible
- Usually has brand name on case

Components on alternator:

- **Pulley:** Belt connection
- **Voltage regulator:** Usually internal (not visible)
- **Output terminal:** Large wire to battery (B+)
- **Field/excitation terminal:** Smaller wire(s)
- **Mounting bracket:** Bolted to engine

Serpentine belt:

- Connects crankshaft pulley to alternator (and other accessories)
- Trace belt path
- Note automatic tensioner (if present)

Battery warning light:

- Dashboard instrument cluster
- Symbol of battery
- Should light with key on, engine off
- Goes out when engine starts

Wiring:

- Heavy output wire (B+)
- Field wires
- Ground connection
- Draw charging system diagram
- Trace power flow from alternator to battery

Pass Criteria:

- ✅ Can locate starter motor
- ✅ Can identify starter cables and wiring
- ✅ Can locate alternator
- ✅ Can identify drive belt
- ✅ Can trace starting system current flow
- ✅ Can trace charging system current flow
- ✅ Creates system diagrams

Hour 2 (2:00 PM - 3:00 PM): Starting System Tests

Activity 1: Voltage Drop Test on Starter Circuit (25 min)

Why voltage drop matters:

- High resistance = voltage lost in cables/connections
- Less voltage reaches starter
- Slow cranking or no cranking

Testing voltage drop (positive side):

1. Set multimeter to DC volts
2. Connect black probe to battery + terminal
3. Connect red probe to starter + terminal (or solenoid B+)
4. Have helper crank engine (brief!)
5. Read voltage while cranking
6. Should be < 0.5V
7. Higher = excessive drop (clean connections or replace cable)

Testing voltage drop (ground side):

1. Black probe to battery - terminal
2. Red probe to starter housing (ground)
3. Crank engine briefly
4. Should be < 0.5V
5. Higher = poor ground

Students practice:

- Test on multiple vehicles
- Record voltage drop readings
- Identify high resistance issues

Activity 2: Starter Current Draw Test (Awareness) (15 min)

- **Requires special equipment:** Amp clamp or carbon pile tester
- **NOT performed in fundamentals** (Electrical Track skill)
- **Understanding the concept:**
  - Normal starter draw: 150-250A (varies by engine)
  - Too high (>300A):
    - Worn starter
    - Engine mechanical problem
    - Dragging problem

  - Too low (<100A):
    - Poor connections
    - Voltage drop issue
  - Actual testing requires special tools

Instructor demonstrates (if equipment available):

- Shows amp clamp on battery cable
- Shows reading during crank
- Explains interpretation

Activity 3: Starting Problem Diagnosis Practice (20 min)

- **Scenario-based learning:**
  - Instructor describes symptoms
  - Students diagnose likely cause
  - Students explain test procedure

Scenarios:

1. "Click, no crank" → Check battery voltage, connections
2. "No click, no crank" → Check battery, neutral switch, ignition switch
3. "Slow cranking" → Check battery, cables, starter condition
4. "Grinding noise" → Starter drive or flywheel teeth worn
5. "Starter stays engaged" → Stuck solenoid

Pass Criteria:

- ✅ Can perform voltage drop test
- ✅ Interprets voltage drop readings
- ✅ Understands current draw testing concept
- ✅ Can diagnose starting problems from symptoms

Hour 3 (3:00 PM - 4:00 PM): Charging System Tests

Activity 1: Charging System Output Test (25 min)

- **Review voltage test** (practiced Day 19):
  - Engine off: ~12.6V
  - Engine running: 13.5-14.5V

Advanced output test:

1. Start engine, let idle
2. Connect multimeter to battery
3. Record voltage (should be 13.5-14.5V)
4. Turn on high loads:
   - Headlights (high beam)
   - Blower fan (high speed)
   - Rear defrost
   - A/C compressor
5. Increase engine speed to 2000 RPM
6. Read voltage
7. Should maintain at least 13.0V
8. If drops below 13.0V = alternator insufficient output

Students practice:

- Test on multiple vehicles
- Record voltage under load
- Determine charging system adequacy

Activity 2: Belt Inspection & Tension Check (20 min) Belt visual inspection:

- **Surface condition:**
  - Cracks across ribs (age wear)
  - Missing chunks
  - Fraying edges
  - Glazed (shiny, slippery surface)
  - Oil contamination

Belt tension check:

- **Modern serpentine belts:** Automatic tensioner
  - Check tensioner position
  - Should be in operational range (marks on tensioner)
  - Beyond marks = belt stretched, replace
- **Older V-belts:** Manual adjustment
  - Press belt midway between pulleys
  - Should deflect 1 cm with firm thumb pressure
  - More = too loose (slipping)
  - Less = too tight (bearing wear)

Tensioner inspection:

- Smooth operation (no binding)

- Spring tension adequate
- Bearing not noisy
- Mounting secure

Students practice:

- Inspect belts on multiple vehicles
- Check tension (if accessible)
- Determine: Good / Replace soon / Replace now

Activity 3: Alternator Problem Diagnosis (15 min)

- **Listen for noises:**
  - Start engine
  - Listen to alternator area
  - **Grinding:** Worn bearing
  - **Whining:** Possible bad diode
  - **Squealing:** Belt slipping

Check battery warning light:

- Key ON, engine OFF: Light should be ON
- Engine running: Light should be OFF
- Light stays ON when running = charging problem

Physical inspection:

- Alternator mounting secure
- No burning smell
- Connections tight
- No damaged wiring

Practice scenarios:

- "Battery light stays on" → Check voltage, belt, alternator
- "Voltage 15.2V" → Overcharging, regulator failure
- "Squealing noise from engine bay" → Belt slipping or worn
- "Voltage 12.8V running" → Not charging, check belt and alternator

Pass Criteria:

- ✅ Can test charging system output
- ✅ Can test voltage under load
- ✅ Interprets voltage readings correctly
- ✅ Can inspect and check belt tension

- ✅ Can identify alternator problems from symptoms

DAY 21 ASSESSMENT:

Practical Skills Checklist:

- [ ] Locates starter motor and components
- [ ] Locates alternator and components
- [ ] Performs voltage drop test on starter circuit
- [ ] Tests charging system output voltage
- [ ] Tests voltage under load
- [ ] Inspects drive belt condition
- [ ] Checks belt tension
- [ ] Diagnoses starting problems from symptoms
- [ ] Diagnoses charging problems from symptoms

Knowledge Check (Verbal):

- What does the starter solenoid do?
- What is normal charging system voltage?
- What causes slow cranking?
- What does battery warning light indicate?
- What happens if alternator overcharges?

DAY 22: WIRING SYSTEMS + WIRE IDENTIFICATION (NO REPAIR)

Learning Objectives:

- Understand automotive wiring harness design
- Identify wire color codes and standards
- Read basic wiring diagrams
- Locate connectors and components
- **NOTE: Identification ONLY - no repair work**

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Automotive Wiring Basics

Purpose of Wiring Harness:

- Distribute electrical power throughout vehicle
- Connect all electrical components
- Provide control signals to modules
- Ground return path

Wire Construction:

- **Conductor:** Copper strands (not solid)
  - Stranded = flexible
  - Multi-strand = more flexible, more expensive
- **Insulation:** PVC, polyethylene, or cross-linked
  - Color-coded for identification
  - Protects from shorts
  - Resists heat, oil, abrasion

Wire Gauge (Thickness):

- **AWG (American Wire Gauge) system**
- **Smaller number = thicker wire = more current capacity**
- Common automotive sizes:
  - 18 AWG: Small loads (indicators, sensors) 5A
  - 16 AWG: Medium loads (lights) 10A
  - 14 AWG: Medium-high loads (blower motor) 15A
  - 12 AWG: High loads (starter signal) 20A
  - 10 AWG: Very high loads (alternator output) 30A
  - 4-6 AWG: Extreme loads (starter motor) 100A+

Why wire size matters:

- Too small = overheating = fire risk
- Too large = expensive, bulky, hard to route
- Must match current requirement

Wiring Harness Types:

- **Main harness:** Engine bay to cabin to rear
- **Engine harness:** All engine-related wiring
- **Dashboard harness:** All interior wiring
- **Door harnesses:** Each door (power windows, locks, speakers)
- **Separate harnesses:** Specific systems (trailer, audio upgrade)

Wiring Protection:

- **Loom/conduit:** Covers and protects wiring
- **Tape:** Wraps and bundles wires
- **Grommets:** Rubber seals where harness passes through body
- **Heat shields:** Protect from exhaust heat
- **Zip ties/clips:** Secure harness to prevent rubbing

Hour 2 (10:00 AM - 11:00 AM): Wire Color Codes & Identification

Color Coding Standards:

- **Primary color:** Base insulation color
- **Tracer/stripe:** Secondary color (thin stripe)
- **Example:** "Green/White" = Green wire with white stripe
  - Sometimes written: "G/W"

Common Color Code Meanings (General Guidelines):

- **Red:** Power/Positive (direct from battery or fused)
- **Black:** Ground/Negative (but not always!)
- **Yellow:** Often switched power (ignition-controlled)
- **Blue:** Often headlights or accessories
- **Green:** Often sensors or data
- **Brown:** Often ground in some systems
- **White:** Often ground, sometimes sensor signal
- **Orange:** Often accessories or switched power
- **Purple/Violet:** Often accessories

IMPORTANT: No universal standard!

- Different manufacturers use different schemes
- Even same manufacturer varies by year/model
- **Always consult wiring diagram for specific vehicle**

Tracer Colors (Stripes):

- Distinguish multiple wires of same primary color
- Example: Blue/Red, Blue/White, Blue/Yellow
- All might be different circuits related to lighting

Reading Wire Labels:

- **Wire size:** 18 AWG, 0.5mm², etc.

- **Color code:** R/W (Red with White stripe)
- **Circuit designation:** Often printed on wire
- **Connector position:** Pin number in connector

Wire Bundles:

- Multiple wires wrapped together
- Protected in loom
- Routed as single harness
- Branches split off to individual components

Connector Types:

1. Blade/Spade Connectors:

- Simple push-on connectors
- Male spade, female receptacle
- Insulated or non-insulated

2. Bullet Connectors:

- Round plug and socket
- Quick disconnect
- Common for accessories

3. Multi-pin Connectors:

- Most common in modern vehicles
- Plastic housing with multiple pins
- 2-pin to 100+ pin connectors
- **Locking tabs:** Secure connection
- **Weather seals:** Keep moisture out
- **Pin assignments:** Specific position for each wire

4. Deutsch Connectors:

- Heavy-duty, weatherproof
- Common in harsh environments
- Screw-lock or bayonet lock

Connector Features:

- **Locking mechanism:** Tab, clip, or bolt
- **Seals:** O-rings or rubber boots
- **Pin retention:** Clips hold pins in housing

- **Color coding:** Connector color may indicate system
- **Keying:** Shaped so can only connect one way

Hour 3 (11:00 AM - 12:00 PM): Reading Wiring Diagrams (Basics)

Why Wiring Diagrams Needed:

- Locate specific wires
- Trace circuit paths
- Understand system operation
- Diagnose electrical problems
- **Essential for professional diagnosis**

Diagram Symbols (Common):

- **Battery:** Two parallel lines (long = +, short = -)
- **Ground:** Multiple lines or triangle pointing down
- **Fuse:** Zigzag line or box with rating
- **Switch:** Break in line with actuator
- **Bulb:** Circle with filament symbol
- **Relay:** Box with coil and contacts
- **Resistor:** Zigzag box
- **Diode:** Arrow with line (current flows one direction)
- **Connector:** Dots or squares where wires meet
- **Splice:** Dot where wires join

Reading a Simple Circuit:

```
[Battery +] --fuse-- --switch-- --load (bulb)-- [Ground]
```

1. Start at power source (battery)
2. Follow wire through protection (fuse)
3. Through control (switch)
4. To load (bulb)
5. Back to ground (completes circuit)

Understanding Connector Pin-Outs:

- Connectors shown with pin numbers
- Table shows which pin is which wire/function
- Example:

```
Connector C123:Pin 1: Red – 12V powerPin 2: Black – GroundPin 3: Blue
– Signal output
```

Color Codes on Diagrams:

- Wires labeled with color abbreviations
- Example: "R/W" = Red with White stripe
- Helps identify physical wire when looking at vehicle

Circuit Numbering:

- Each circuit has designation
- Example: "Circuit 14 - Headlight Low Beam"
- Helps organize complex diagrams

Diagram Types (Awareness):

- **Schematic:** Shows circuit operation
- **Wiring harness:** Shows physical routing
- **Component location:** Shows where things are
- **Connector pin-out:** Shows pin assignments

Using Diagrams for Troubleshooting:

- Locate non-working component on diagram
- Trace back to power source
- Identify fuses, relays, switches in path
- Test each point systematically
- **Detailed troubleshooting = Diagnostics Track**

NOTE: Fundamentals students learn to READ diagrams, not advanced diagnosis

---

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Wire and Connector Identification

Activity 1: Wire Color Identification Practice (25 min)

- Students examine actual vehicle wiring:
  - Select accessible harness (under hood, door jamb)
  - Identify individual wire colors
  - Record: Primary color, tracer color (if any)

  - Use notation: "Red/White" or "R/W"

Practice identifying wires:

- Find and document at least 10 different wires
- Create wire identification chart:

```
Wire 1: Red – likely power
Wire 2: Black – likely ground
Wire 3: Blue/White – unknown function
Wire 4: Green – likely sensor
...etc.
```

- Compare findings with wiring diagram (if available)

Activity 2: Connector Identification (20 min)

- Locate different connector types on vehicle:
  - **Under hood:** Engine sensors, injectors
  - **Behind dashboard:** Module connectors
  - **Doors:** Power window/lock connectors
  - **Lights:** Bulb sockets

For each connector:

- Identify type (blade, multi-pin, etc.)
- Count pins (2-pin, 4-pin, etc.)
- Identify locking mechanism
- Note color (if coded)
- Check condition (corrosion, damage)

Activity 3: Following a Wire Path (15 min)

- **Trace exercise:**
  - Instructor selects a specific circuit (e.g., horn)
  - Students physically trace wire from:
    - Component (horn)
    - Back through harness
    - To fuse box
    - To relay (if present)
    - To switch (steering wheel)
- Note how harness branches and routes
- Understand physical wire paths
- Appreciate harness complexity

Pass Criteria:

- ✅ Can identify wire colors correctly
- ✅ Can use color code notation
- ✅ Can identify different connector types
- ✅ Can trace wire paths physically
- ✅ Recognizes harness routing

Hour 2 (2:00 PM - 3:00 PM): Connector Inspection & Wiring Diagram Practice

Activity 1: Connector Condition Inspection (25 min)

- **Systematic connector inspection:**

Visual checks:

- **Connector housing:**
  - Cracks, damage
  - Locking tab intact
  - Proper seating (fully inserted)
- **Pins/terminals:**
  - Corrosion (green/white deposits)
  - Bent pins
  - Pushed-back pins (not fully seated)
  - Burned/melted terminals
- **Seals:**
  - Present and intact
  - Not torn or deteriorated
- **Wires at connector:**
  - Not pulling out
  - Insulation intact up to connector
  - No chafing

Common connector problems:

- Corrosion (most common)
  - Caused by moisture intrusion
  - Increases resistance
  - Component works intermittently
- Loose connection
  - Vibration separates connector

    - Intermittent operation
- Damaged pins
    - Bent or broken
    - Pushed back in housing
    - No contact when mated

Students inspect multiple connectors:

- Document condition (Good / Monitor / Service needed)
- Take photos of problems
- Create inspection report

IMPORTANT NOTE: Students are inspecting only, NOT repairing connectors

- Connector repair = Electrical Track skill
- Fundamentals = identification and understanding

Activity 2: Wiring Diagram Reading Practice (35 min)

- Instructor provides simple wiring diagrams
- Students practice reading:

Exercise 1: Headlight circuit

- Trace power from battery
- Through fuse
- Through switch
- To headlight
- Identify wire colors at each point
- Understand complete path

Exercise 2: Horn circuit

- Find horn relay on diagram
- Trace trigger from steering wheel button
- Trace power path through relay
- To horn
- Understand relay control vs power circuit

Exercise 3: Cooling fan circuit

- Identify temperature switch/sensor
- Find cooling fan relay
- Trace ECU control (if present)
- Understand fan activation logic

For each diagram:

- Identify components
- Follow wire paths
- Note wire colors
- Understand circuit operation
- Locate physical components on vehicle (compare diagram to reality)

Pass Criteria:

- ✅ Can inspect connector condition systematically
- ✅ Identifies common connector problems
- ✅ Can read basic wiring diagrams
- ✅ Can trace circuits on diagrams
- ✅ Can correlate diagram to physical vehicle

Hour 3 (3:00 PM - 4:00 PM): Electrical System Problem Awareness

Activity 1: Common Wiring Problems (Discussion) (20 min)

Problem Categories:

1. Corrosion:

- **Where:** Connectors, especially exposed to elements
- **Causes:** Moisture, salt, age
- **Symptoms:**
  - Intermittent operation
  - Component works sometimes
  - Wiggling connector makes it work
- **Identification:** Green/white powder on pins

2. Chafing/Abrasion:

- **Where:** Wires rubbing against metal edges
- **Causes:** Poor routing, missing grommets, vibration
- **Symptoms:**
  - Blown fuses
  - Intermittent shorts
  - Burning smell
- **Identification:** Worn insulation, exposed copper

3. Broken Wires:

- **Where:** Flex points (door hinges, hood)
- **Causes:** Repeated flexing, pulling, age
- **Symptoms:**
  - Complete loss of function
  - May be intermittent at break point
- **Identification:** Wire broken inside insulation (not always visible)

4. Rodent Damage:

- **Where:** Anywhere, especially parked vehicles
- **Causes:** Rodents chew soy-based insulation
- **Symptoms:** Multiple circuits fail, blown fuses
- **Identification:** Chewed insulation, nesting material

5. Poor Grounds:

- **Where:** Ground connection points
- **Causes:** Corrosion, loose bolts, paint blocking connection
- **Symptoms:**
  - Dim lights
  - Slow operation
  - Multiple systems affected
- **Identification:** Voltage present but component doesn't work

Activity 2: When to Refer to Electrical Specialist (15 min)

- **Fundamentals technicians can:**
  - ✅ Identify problems visually
  - ✅ Test voltage and continuity
  - ✅ Replace fuses
  - ✅ Clean connections (basic)
  - ✅ Tighten connections
- **Refer to Electrical Track specialist for:**
  - ❌ Wire repair (splicing, soldering)
  - ❌ Connector repair (pin replacement)
  - ❌ Harness fabrication
  - ❌ Complex diagnosis (multiple system faults)
  - ❌ Module programming/coding
  - ❌ CAN bus problems
  - ❌ Parasitic drain diagnosis

Understanding your limits:

- Attempting repairs without training can:
  - Make problem worse
  - Create safety hazards
  - Void warranties
  - Cause expensive damage
- Better to recognize and refer than attempt and fail

Activity 3: Electrical Safety Review (25 min)

Working Safely with Vehicle Wiring:

1. Battery Safety:

- Disconnect battery before working on wiring (in most cases)
- Prevents shorts
- Prevents airbag deployment
- Wait 10 minutes after disconnecting (capacitors discharge)

2. Avoid Shorts:

- Never allow bare wires to touch ground
- Never probe connectors randomly (can damage modules)
- Use proper tools (insulated)
- One circuit at a time

3. Fuse Replacement Safety:

- **Always use correct amperage** (never higher!)
- Using higher fuse = fire risk
- If fuse blows repeatedly = find and fix cause (don't keep replacing)

4. Working Around Airbags:

- Yellow connectors = airbag circuits
- Never probe or disconnect without proper procedure
- Accidental deployment = injury + expensive
- See Day 17 safety rules

5. High Voltage Safety (Hybrid/EV):

- Orange cables = high voltage (200-600V)
- **Never touch without proper training**
- Can be lethal

- Requires special certification
- Not covered in fundamentals

6. Personal Safety:

- Remove jewelry (conducts electricity)
- Use insulated tools
- Safety glasses (arc flash risk)
- Don't work on live circuits if avoidable
- Have fire extinguisher nearby

Students review and sign electrical safety acknowledgment:

- Understands battery safety
- Knows proper fuse usage
- Recognizes limits of skills
- Will refer complex problems
- Follows safety procedures

Pass Criteria:

- ✅ Can identify common wiring problems
- ✅ Understands when to refer to specialist
- ✅ Follows electrical safety procedures
- ✅ Knows limitations of fundamental skills
- ✅ Uses proper safety practices

DAY 22 ASSESSMENT:

Practical Skills Checklist:

- [ ] Identifies wire colors correctly
- [ ] Uses color code notation properly
- [ ] Identifies different connector types
- [ ] Inspects connector condition
- [ ] Traces wire paths physically
- [ ] Reads basic wiring diagrams
- [ ] Identifies common wiring problems
- [ ] Follows electrical safety procedures
- [ ] Knows when to refer problems

Knowledge Check (Verbal):

- What does smaller AWG number mean?
- How do you read wire color code "R/W"?
- What are three common connector problems?
- Why must we use correct fuse amperage?
- When should battery be disconnected?

DAY 23: SENSORS & ACTUATORS + COMPONENT IDENTIFICATION

Learning Objectives:

- Understand sensor and actuator purposes (awareness level)
- Identify common sensors on vehicle
- Understand basic sensor operation principles
- Locate actuators and understand their functions
- **NOTE: Identification ONLY - no testing or diagnosis**

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): Sensor Fundamentals

What are Sensors?

- **Input devices to ECU (Engine Control Unit)**
- Convert physical conditions to electrical signals
- Tell ECU what's happening in real-time
- **"Eyes and ears" of the engine management system**

Why Sensors are Critical:

- Modern engines controlled by computer (ECU)
- ECU needs information to make decisions:
  - How much fuel to inject
  - When to fire spark plugs
  - When to shift transmission
  - When to activate safety systems
- Faulty sensor = incorrect information = poor performance

Basic Sensor Types by Output Signal:

1. Voltage Generators (Active Sensors):

- Produce their own voltage
- Don't need external power
- **Examples:**
  - Oxygen sensor (generates voltage based on oxygen)
  - Magnetic sensors (generate voltage from magnetism)

2. Variable Resistance (Passive Sensors):

- Resistance changes with condition
- Need reference voltage from ECU
- ECU reads voltage change
- **Examples:**
  - Temperature sensors (resistance changes with heat)
  - Throttle position sensor (resistance changes with position)

3. On/Off Switches:

- Simple open/closed circuit
- Binary information (yes/no)
- **Examples:**
  - Brake light switch
  - Clutch position switch
  - Pressure switches

4. Frequency/Digital Sensors:

- Output pulses or digital signals
- Frequency changes with speed/position
- **Examples:**
  - Wheel speed sensors (ABS)
  - Crankshaft position sensor
  - Vehicle speed sensor

Sensor Signal Types:

- **Analog:** Continuous varying voltage (0-5V)
  - Example: Throttle position 0-100% = 0.5-4.5V
- **Digital:** On/off pulses
  - Example: Crankshaft sensor pulses
- **Frequency:** Varying pulse rate
  - Example: Vehicle speed sensor (more pulses = faster speed)

Common Sensor Problems:

- **Open circuit:** Broken wire, failed sensor (no signal)
- **Short circuit:** Wire touching ground (incorrect signal)
- **Out of range:** Signal too high or too low
- **Drift:** Sensor reads incorrectly (needs calibration or replacement)
- **Intermittent:** Works sometimes (loose connection, heat-related)

Hour 2 (10:00 AM - 11:00 AM): Common Automotive Sensors

Engine Management Sensors:

1. Mass Airflow (MAF) Sensor:

- **Location:** Between air filter and throttle body
- **Purpose:** Measures amount of air entering engine
- **How it works:**
  - Hot wire or hot film element
  - Air flow cools element
  - More cooling = more current needed to maintain temperature
  - ECU reads current = knows airflow
- **ECU uses for:** Calculating fuel injection amount
- **Failure symptoms:** Poor fuel economy, hesitation, black smoke

2. Manifold Absolute Pressure (MAP) Sensor:

- **Location:** On intake manifold
- **Purpose:** Measures intake manifold pressure/vacuum
- **How it works:**
  - Pressure/vacuum moves diaphragm
  - Changes electrical resistance or capacitance
  - ECU reads voltage = knows engine load
- **ECU uses for:** Calculating fuel and ignition timing
- **Alternative to MAF** (some vehicles use MAP instead)

3. Throttle Position Sensor (TPS):

- **Location:** On throttle body, connected to throttle plate shaft
- **Purpose:** Tells ECU throttle opening percentage
- **How it works:**
  - Potentiometer (variable resistor)
  - Rotates with throttle shaft

  - 0% throttle = ~0.5V
  - 100% throttle = ~4.5V
- **ECU uses for:** Acceleration enrichment, shift points
- **Failure symptoms:** Poor acceleration, rough idle, transmission issues

4. Coolant Temperature Sensor (CTS/ECT):

- **Location:** In engine coolant passage (thermostat housing or head)
- **Purpose:** Measures engine temperature
- **How it works:**
  - Thermistor (resistance changes with temperature)
  - Cold engine = high resistance = high voltage
  - Hot engine = low resistance = low voltage
- **ECU uses for:**
  - Fuel enrichment when cold
  - Fan control
  - Transmission shift points
- **Failure symptoms:** Hard starting, poor cold running, overheating

5. Intake Air Temperature (IAT) Sensor:

- **Location:** In air filter housing or MAF sensor
- **Purpose:** Measures incoming air temperature
- **How it works:** Same as coolant sensor (thermistor)
- **ECU uses for:** Fuel calculation (cold air denser = more fuel needed)

6. Oxygen Sensor (O2 / Lambda Sensor):

- **Location:** Exhaust manifold (before catalyst) and after catalyst
- **Purpose:** Measures oxygen in exhaust gases
- **How it works:**
  - Zirconia element generates voltage
  - Low oxygen (rich mixture) = high voltage (~0.9V)
  - High oxygen (lean mixture) = low voltage (~0.1V)
  - Switches rapidly between rich/lean at ideal mixture
- **ECU uses for:** Fuel trim (fine-tuning air/fuel ratio)
- **Types:**
  - Narrowband (switching type, older)
  - Wideband (linear, more precise, modern)
- **Heated:** Has internal heater for faster operation

7. Crankshaft Position Sensor (CKP):

- **Location:** Near crankshaft pulley or flywheel
- **Purpose:** Tells ECU crankshaft position and RPM
- **How it works:**
  - Magnetic sensor reads teeth on reluctor wheel
  - Missing tooth indicates TDC (Top Dead Center)
  - Produces AC voltage pulses
- **ECU uses for:**
  - Ignition timing
  - Fuel injection timing
  - RPM calculation
- **Critical:** Engine won't start if failed

8. Camshaft Position Sensor (CMP):

- **Location:** On cylinder head near camshaft
- **Purpose:** Tells ECU which cylinder is firing
- **How it works:** Similar to CKP (magnetic or Hall effect)
- **ECU uses for:**
  - Sequential fuel injection
  - Ignition timing
  - Variable valve timing control
- **Failure:** May run rough, may not start (some vehicles)

9. Knock Sensor:

- **Location:** On engine block
- **Purpose:** Detects engine knock/detonation
- **How it works:**
  - Piezoelectric (vibration creates voltage)
  - Knock creates specific frequency vibration
- **ECU uses for:** Retard timing to prevent damage
- **Failure:** Reduced performance, potential engine damage

Hour 3 (11:00 AM - 12:00 PM): Other Sensors & Actuators

Chassis/Safety Sensors:

10. Wheel Speed Sensors (ABS):

- **Location:** At each wheel hub
- **Purpose:** Measure individual wheel speeds

- **How it works:**
  - Magnetic sensor reads toothed ring (tone ring)
  - Produces frequency signal (more pulses = faster)
- **Used for:** ABS, traction control, stability control
- **Failure:** ABS light on, ABS disabled

11. Steering Angle Sensor:

- **Location:** In steering column
- **Purpose:** Measures steering wheel position and rate
- **Used for:** Stability control, lane keeping assist
- **Failure:** Stability control light, systems disabled

12. Brake Pedal Position Sensor:

- **Location:** At brake pedal
- **Purpose:** Detects when brake pedal pressed
- **Used for:** Brake lights, cruise cancel, start-stop systems

Transmission Sensors:

13. Transmission Input/Output Speed Sensors:

- **Purpose:** Measure transmission shaft speeds
- **Used for:** Shift point calculation, gear ratio verification
- **Failure:** Transmission stuck in one gear, harsh shifting

14. Transmission Fluid Temperature Sensor:

- **Purpose:** Monitor ATF temperature
- **Used for:** Protect transmission from overheating

Climate Control Sensors:

15. Ambient Temperature Sensor:

- **Location:** Behind front bumper
- **Purpose:** Measure outside air temperature
- **Used for:** Climate control, ice warning

16. Cabin Temperature Sensor:

- **Location:** In dashboard
- **Purpose:** Measure cabin temperature
- **Used for:** Automatic climate control

Actuators - Output Devices:

What are Actuators?

- **Output devices controlled by ECU**
- Convert electrical signals to mechanical action
- **"Muscles" of the vehicle systems**
- ECU commands them based on sensor inputs

Common Actuators:

1. Fuel Injectors:

- **Purpose:** Spray precise amount of fuel
- **How it works:**
  - Solenoid-operated valve
  - ECU pulses open (milliseconds)
  - Longer pulse = more fuel
- **Control:** Pulse width modulation (PWM)

2. Ignition Coils:

- **Purpose:** Create high voltage for spark plugs
- **Control:** ECU grounds coil at precise timing

3. Idle Air Control (IAC) Valve:

- **Purpose:** Control idle speed
- **How it works:**
  - Stepper motor or solenoid
  - Opens/closes bypass around throttle
  - More air = higher idle

4. EVAP Purge Valve:

- **Purpose:** Release fuel vapors to engine
- **Control:** ECU opens when conditions right

5. EGR Valve:

- **Purpose:** Recirculate exhaust gases
- **Types:** Vacuum-operated or electronic
- **Control:** ECU commands based on conditions

6. VVT Solenoid (Variable Valve Timing):

- **Purpose:** Adjust camshaft timing
- **Control:** ECU-controlled oil pressure valve
- **Effect:** Optimize performance/economy

7. Cooling Fan Relay:

- **Purpose:** Turn radiator fan on/off
- **Control:** ECU grounds relay coil

8. A/C Compressor Clutch:

- **Purpose:** Engage/disengage A/C compressor
- **Control:** ECU or climate control module

9. Transmission Solenoids:

- **Purpose:** Control hydraulic flow for gear changes
- **Control:** ECU pulses to shift gears

Actuator Testing (Awareness):

- Can hear/feel operation (clicks, movement)
- Advanced testing requires scan tool (bidirectional control)
- **Bidirectional testing = Diagnostics Track**
- Fundamentals = identification only

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Sensor Location & Identification

Activity 1: Engine Management Sensor Hunt (40 min)

- Students locate and identify on actual vehicle:

Air Intake Sensors:

- **MAF sensor:** Between air filter and throttle body
  - Usually black box with electrical connector
  - Visible hot wire or element inside
- **Or MAP sensor:** On intake manifold
  - Small sensor with vacuum connection
- **IAT sensor:** May be integrated in MAF or separate

- **Throttle Position Sensor:** On throttle body
  - Rotates with throttle shaft

Temperature Sensors:

- **Coolant temperature sensor:** On engine (thermostat housing or head)
  - Single or dual wire connector
  - Threaded into coolant passage
- **Oil temperature sensor (if equipped):** On oil filter housing or pan

Position/Speed Sensors:

- **Crankshaft position sensor:** Near crank pulley or flywheel
  - 2 or 3 wire connector
  - Small gap to reluctor wheel
- **Camshaft position sensor:** On cylinder head
  - Similar appearance to CKP
- **Vehicle speed sensor:** On transmission
  - May be integrated with output speed sensor

Exhaust Sensors:

- **Oxygen sensors:** In exhaust
  - Before catalyst (upstream) - usually 1 or 2
  - After catalyst (downstream) - usually 1 or 2
  - 1, 2, 3, or 4 wires (heated sensors have more)

For each sensor found:

- Take photo
- Note location precisely
- Count wires in connector
- Identify connector type
- Create sensor location map/diagram

Activity 2: Chassis Sensor Identification (20 min)

- **Wheel speed sensors:** At each wheel hub
  - Small sensor near brake rotor or in hub
  - Wire runs to ABS module
  - One per wheel (4 total)
- **Brake light switch:** At brake pedal
  - Activates when pedal pressed
- **Clutch position switch (manual):** At clutch pedal

    - Prevents starting unless clutch depressed
- **Transmission speed sensors:** On transmission case
    - Input and output speed sensors
- Add chassis sensors to location map

Pass Criteria:

- ✅ Can locate MAF or MAP sensor
- ✅ Can locate throttle position sensor
- ✅ Can locate coolant temperature sensor
- ✅ Can locate crankshaft position sensor
- ✅ Can locate oxygen sensors
- ✅ Can locate wheel speed sensors
- ✅ Creates comprehensive sensor location map

Hour 2 (2:00 PM - 3:00 PM): Actuator Location & Identification

Activity 1: Engine Actuator Hunt (35 min)

- Students locate and identify actuators:

Fuel System:

- **Fuel injectors:** On fuel rail or intake manifold
    - One per cylinder
    - Electrical connector on each
    - Makes clicking sound when operating

Ignition System:

- **Ignition coils:** On or near spark plugs
    - One coil per cylinder (most modern)
    - Or single coil with distributor (older)

Idle/Air Control:

- **Idle air control valve:** On throttle body or intake manifold
    - Stepper motor or solenoid
    - Controls bypass air

Emissions:

- **EVAP purge valve:** Near intake manifold

    - Solenoid valve
    - Vacuum hoses connected
- **EGR valve:** On intake manifold or cylinder head
    - Larger than most sensors
    - May have vacuum or electrical control

Variable Systems:

- **VVT solenoid(s):** On cylinder head
    - Near camshaft(s)
    - Oil pressure control valve

For each actuator:

- Photograph
- Note electrical connector
- Understand basic function
- Add to location map

Activity 2: Cooling & A/C Actuators (15 min)

- **Cooling fan:** Behind radiator
    - Electric motor
    - Relay-controlled
- **Thermostat:** Engine coolant passage
    - Traditional: Mechanical (wax pellet)
    - Modern: May have electronic control
- **A/C compressor clutch:** On compressor
    - Electromagnetic clutch
    - Engages compressor pulley

Activity 3: Transmission Actuators (10 min)

- **Transmission solenoids:** Inside transmission
    - Not visible without disassembly
    - Multiple solenoids control shifting
    - Connector visible on transmission case

Pass Criteria:

- ✅ Can locate fuel injectors
- ✅ Can locate ignition coils
- ✅ Can locate EVAP purge valve
- ✅ Can locate EGR valve (if equipped)

- ✅ Can identify cooling fan
- ✅ Adds actuators to location map

Hour 3 (3:00 PM - 4:00 PM): Sensor/Actuator Operation Observation

Activity 1: Listening to Actuators (25 min)

- **With engine running, listen for:**

Fuel injectors:

- Place screwdriver against injector (handle to ear)
- Hear rapid clicking (pulsing)
- All cylinders should sound similar
- Demonstrates they're operating

Idle air control:

- Engine at idle
- May hear slight hiss of air
- Some have audible motor movement

Cooling fan:

- When fan runs, obvious motor noise
- May cycle on/off based on temperature

EVAP purge valve:

- May hear clicking when activated
- Usually when engine at operating temp

Activity 2: Visual Operation Checks (20 min)

- **Things you can see operating:**

Throttle position:

- Have assistant press accelerator slowly
- Watch throttle plate open
- TPS rotates with shaft (can't see signal, but understand relationship)

VVT operation:

- Can't see directly (internal)

- But understand it operates based on conditions

Ignition coils:

- Can see spark (with timing light or in dark)
- **DO NOT TOUCH when engine running!** (high voltage)

Oxygen sensors:

- Heat up when running (heated sensor)
- Can't see signal without scan tool

Activity 3: Understanding Sensor-to-Actuator Relationships (15 min)

- **Discussion: How sensors and actuators work together**

Example 1: Acceleration

1. Driver presses pedal
2. **TPS sensor** signals throttle opening
3. **MAF/MAP sensor** measures more air entering
4. ECU calculates fuel needed
5. ECU commands **fuel injectors** to pulse longer
6. More fuel injected
7. **Oxygen sensors** verify mixture correct
8. ECU adjusts **injectors** for perfect ratio

Example 2: Cold Start

1. **Coolant temp sensor** reads cold (10°C)
2. ECU knows engine needs more fuel
3. ECU pulses **injectors** longer (rich mixture)
4. ECU may increase **idle air control** (faster idle)
5. As engine warms, ECU gradually reduces enrichment
6. **Oxygen sensors** take over when warmed up

Example 3: Cruise Control

1. Driver sets cruise at 100 km/h
2. **Vehicle speed sensor** measures current speed
3. Going uphill, speed drops to 95 km/h
4. ECU opens throttle via **throttle actuator** (electronic throttle)
5. Or on cable throttle: Pulls cable via servo motor
6. Speed increases back to 100 km/h
7. ECU maintains speed by adjusting throttle

Students create sensor-actuator relationship diagrams:

- Choose a scenario (starting, accelerating, etc.)
- Draw which sensors provide input
- Draw which actuators receive commands
- Show ECU in middle processing information

Pass Criteria:

- ✅ Can observe actuator operation
- ✅ Can listen for actuator sounds
- ✅ Understands sensor-to-actuator relationships
- ✅ Can explain basic control loops
- ✅ Creates relationship diagrams

DAY 23 ASSESSMENT:

Practical Skills Checklist:

- [ ] Locates major engine sensors (MAF/MAP, TPS, CTS, CKP, O2)
- [ ] Locates chassis sensors (wheel speed, brake switch)
- [ ] Locates fuel injectors
- [ ] Locates ignition coils
- [ ] Locates emission actuators (EVAP, EGR)
- [ ] Can observe actuator operation
- [ ] Creates sensor/actuator location map
- [ ] Understands sensor-to-actuator relationships

Knowledge Check (Verbal):

- What is the difference between a sensor and an actuator?
- Name three engine management sensors
- What does the oxygen sensor measure?
- What do fuel injectors do?
- How do sensors and actuators work together?

DAY 24: INTRODUCTION TO DIAGNOSTICS + BASIC OBD-II SCANNER OPERATION

Learning Objectives:

- Understand OBD-II system purpose and standards
- Locate OBD-II diagnostic connector
- Connect scanner and perform basic scan
- Read and understand fault codes (basic level)
- View live data (awareness level)
- **NOTE: Basic operation ONLY - advanced diagnosis in Diagnostics Track**

MORNING SESSION: THEORY (3 Hours)

Hour 1 (9:00 AM - 10:00 AM): OBD-II System Overview

What is OBD-II?

- **On-Board Diagnostics, Second Generation**
- Standardized diagnostic system
- **Required in India:** BS4 vehicles onwards (2017+)
- **Monitors:** Emission control systems and related components

Purpose of OBD-II:

- Monitor emission system performance
- Detect malfunctions that increase emissions
- Store fault codes when problems detected
- Illuminate warning light (MIL - Malfunction Indicator Lamp)
- Provide access for diagnostic tools
- Help technicians diagnose problems faster

OBD-II Standards:

- **Diagnostic connector:** 16-pin, standardized location
- **Fault code format:** Standardized (P0xxx, P1xxx, etc.)
- **Communication protocols:** Several types, but standardized interface
- **Readiness monitors:** Verify emission systems tested
- **Freeze frame:** Capture conditions when fault occurred

Check Engine Light (MIL):

- **Yellow/amber engine symbol** on dashboard
- **Solid light:** Problem detected, stored as code
  - Usually drivable
  - Get diagnosed soon
- **Flashing light:** Severe misfire

    - Stop driving immediately
    - Can damage catalytic converter
- **Light off:** No current faults (or faults cleared, but may return)

When MIL Illuminates:

- ECU detects problem
- Runs tests to verify (not one-time glitch)
- Sets fault code
- Turns on light
- Stores freeze frame data
- May turn off if fault doesn't reoccur (code still stored)

Diagnostic Trouble Codes (DTCs):

- 5-character codes identify faults
- Format: **X####**
- **First character - System:**
    - **P** = Powertrain (engine, transmission)
    - **B** = Body (airbags, climate, etc.)
    - **C** = Chassis (ABS, steering, suspension)
    - **U** = Network (communication between modules)
- **Second character - Code Type:**
    - **0** = Generic/SAE (standardized, all manufacturers)
    - **1** = Manufacturer-specific (unique to brand)
- **Third character - Subsystem:**
    - 1 = Fuel/air metering
    - 2 = Fuel/air metering (injector circuit)
    - 3 = Ignition system
    - 4 = Auxiliary emission controls
    - 5 = Idle control
    - 6 = ECU and outputs
    - 7 & 8 = Transmission
- **Last two characters:** Specific fault number

Example Code: P0301

- **P** = Powertrain
- **0** = Generic code
- **3** = Ignition system
- **01** = Cylinder 1 misfire detected

Example Code: P0420

- **P** = Powertrain
- **0** = Generic
- **4** = Auxiliary emission
- **20** = Catalyst system efficiency below threshold (Bank 1)

Hour 2 (10:00 AM - 11:00 AM): OBD-II Components & Communication

OBD-II System Components:

1. ECU (Engine Control Unit):

- Main computer
- Monitors all sensors
- Runs diagnostic tests
- Stores fault codes
- Controls MIL
- Communicates with scan tool

2. Diagnostic Connector (DLC):

- 16-pin connector
- **Standard location:** Under dashboard, driver side
  - Within 300mm of steering wheel centerline
  - Some in center console, some near fuse box
- **Pins:**
  - Pin 16: Battery voltage (+12V)
  - Pin 4 or 5: Ground
  - Pins 2, 6, 7, 10, 14, 15: Communication (protocol-specific)

3. Sensors & Actuators:

- All monitored by ECU
- Failures trigger codes

4. MIL (Malfunction Indicator Lamp):

- Dashboard warning light
- Controlled by ECU

Communication Protocols:

- **Multiple protocols exist:**
  - ISO 9141-2 (older)
  - ISO 14230 (KWP2000)
  - SAE J1850 (GM, Ford older)
  - CAN Bus (most modern, 2008+)
- **Scanner auto-detects protocol** (usually)
- Technician doesn't need to know details
- **CAN = fastest, most reliable**

OBD-II Monitors:

- **Continuous monitors:** Run all the time
  - Misfire detection
  - Fuel system
  - Comprehensive component monitoring
- **Non-continuous monitors:** Run under specific conditions
  - Catalyst efficiency
  - Oxygen sensor
  - EGR system
  - EVAP system
  - Secondary air (if equipped)

Monitor Status (Readiness):

- **Ready/Complete:** Monitor has run and passed
- **Not Ready/Incomplete:** Monitor hasn't run yet
- **Important for emission testing:**
  - Most monitors must be "Ready"
  - After clearing codes or battery disconnect, monitors reset
  - Must drive through "drive cycle" to set monitors

Hour 3 (11:00 AM - 12:00 PM): Using OBD-II Scanner - Theory

Types of Scanners:

1. Basic Code Readers:

- Read and clear codes
- Turn off MIL
- Very basic live data
- Inexpensive (₹2,000-5,000)

- Good for DIY

2. OBD-II Scanners:

- All code reader functions
- More detailed live data
- Freeze frame data
- Monitor status
- Moderate price (₹5,000-20,000)
- **Suitable for fundamentals**

3. Professional Scan Tools:

- All scanner functions
- Bidirectional control
- Module coding/programming
- Manufacturer-specific codes
- Advanced diagnostics
- Expensive (₹50,000-500,000)
- **Used in Diagnostics Track**

Basic Scanner Functions:

1. Read Codes:

- Retrieves stored fault codes
- Shows code and description
- Example: "P0301 - Cylinder 1 Misfire Detected"

2. Clear Codes:

- Erases stored codes
- Turns off MIL
- Resets monitors to "Not Ready"
- **Only clear AFTER repair!**

3. View Freeze Frame:

- Snapshot of conditions when code set
- Shows:
    - RPM at time of fault
    - Coolant temperature
    - Vehicle speed
    - Fuel trim

    - Other parameters
- Helps diagnose intermittent problems

4. View Live Data:

- Real-time sensor values
- Can view multiple parameters
- **Examples:**
    - Engine RPM: 850 RPM (idle)
    - Coolant temp: 85°C
    - Throttle position: 15%
    - O2 sensor: 0.45V (switching)
    - Fuel trim: +2% (normal)
- Helps verify sensor operation

5. Monitor Status:

- Shows which monitors ready/not ready
- Important for emission testing

6. Vehicle Information:

- VIN (Vehicle Identification Number)
- Calibration ID
- Sometimes odometer reading

Understanding What Scanner Shows:

- **Scanner shows codes, not solutions**
- Code indicates symptom or circuit fault
- **Example:** P0420 (Catalyst efficiency low)
    - Could be: Bad catalyst, bad O2 sensor, exhaust leak, engine running rich
    - Requires further diagnosis
- **Codes are starting point, not answer**

When to Scan:

- Check engine light on
- Performance problems
- Failed emission test
- After repairs (verify fix)
- Regular diagnostic check

Scanner Limitations (Basic Units):

- Can't test actuators (bidirectional)
- Can't program modules
- May not show manufacturer codes
- Limited data graphing
- Can't access all modules (body, ABS may require advanced tool)

AFTERNOON SESSION: PRACTICAL (3 Hours)

Hour 1 (1:00 PM - 2:00 PM): Locating DLC & Connecting Scanner

Activity 1: DLC Location Practice (20 min)

- Students locate DLC on multiple vehicles:
  - **Most common:** Under dashboard, left of steering column
  - **Also check:** Center console, near fuse box
  - **Look for:** Trapezoid-shaped 16-pin connector
  - **May have cover:** Plastic cover snaps off

Identifying DLC:

- 16 pins in 2 rows (8 pins each)
- Keyed shape (only fits one way)
- Usually black plastic
- May have label "OBD-II" or "DIAGNOSTICS"

DLC condition check:

- Pins not bent or missing
- No corrosion
- Not damaged
- Secure to bracket (not hanging loose)

Activity 2: Scanner Connection Procedure (25 min)

- **Equipment:** Basic OBD-II scanner

Connection procedure:

1. **Turn ignition OFF**
2. **Locate DLC**
3. **Align scanner connector with DLC**
   - Match keying (only fits one way)
4. **Push firmly until clicks** (fully seated)

5. **Turn ignition to ON position** (engine off or running, depends on scanner)
6. **Scanner powers up** (gets power from DLC pin 16)
7. **Scanner auto-detects vehicle** (reads VIN, protocol)
8. **Wait for scanner ready** (may take 10-30 seconds)

Students practice:

- Each student connects scanner to vehicle
- Powers up successfully
- Observes communication established
- Verifies vehicle info displayed

Activity 3: Scanner Navigation (15 min)

- **Learning scanner menu:**
  - Usually button or touchscreen interface
  - **Main menu options:**
    - Read codes
    - Erase codes
    - Live data
    - Freeze frame
    - Monitor status
    - Vehicle info
    - Settings
- Navigate through menus
- Don't perform functions yet (just familiarize)
- Understand menu structure
- Find help/manual function (if present)

Pass Criteria:

- ✅ Can locate DLC on various vehicles
- ✅ Can connect scanner correctly
- ✅ Achieves communication with vehicle
- ✅ Can navigate scanner menus

Hour 2 (2:00 PM - 3:00 PM): Reading & Understanding Fault Codes

Activity 1: Reading Stored Codes (25 min)

- **Procedure:**

1. Scanner connected and communicating
2. Select "Read Codes" or "Diagnostic Trouble Codes"
3. Scanner retrieves codes from ECU
4. Wait for retrieval (may take few seconds)
5. Scanner displays codes

What scanner shows:

- **Code number:** P0301
- **Description:** Cylinder 1 Misfire Detected
- **Status:** Pending or Confirmed
- **Number of codes:** May be multiple codes

Understanding code status:

- **Pending:** Detected once, not confirmed yet
- **Confirmed/Stored:** Detected multiple times, MIL on
- **Permanent:** Can't be cleared until ECU verifies repair

Students practice:

- Read codes from vehicles (instructor may set codes for practice)
- Record all codes found
- Note code descriptions
- Understand status

Activity 2: Interpreting Common Codes (20 min)

- **Common codes and meanings:**

P0300-P0304: Misfire codes

- P0300 = Random misfire
- P0301 = Cylinder 1 misfire
- P0302 = Cylinder 2 misfire, etc.
- **Possible causes:** Spark plug, coil, injector, compression, fuel

P0420/P0430: Catalyst efficiency

- Catalyst not working effectively
- **Possible causes:** Bad catalyst, O2 sensor, exhaust leak, rich/lean running

P0171/P0174: System too lean

- Not enough fuel or too much air

- **Possible causes:** Vacuum leak, MAF sensor, fuel pressure, injector

P0128: Coolant thermostat

- Engine not reaching operating temperature
- **Possible causes:** Stuck-open thermostat, faulty sensor

P0442: EVAP small leak

- Fuel vapor leak
- **Possible causes:** Loose gas cap (most common!), cracked hose, purge valve

Discussion:

- **Codes indicate problem area, not exact part**
- **Multiple codes may have single root cause**
- **Example:** Bad MAF causes P0171 (lean), P0300 (misfire), P0420 (catalyst)
- **Always verify with further diagnosis**

Activity 3: Freeze Frame Data Review (15 min)

- **Access freeze frame:**
  1. Select code
  2. View freeze frame (if available)
  3. Review conditions when code set

Freeze frame parameters:

- Engine RPM
- Engine load
- Coolant temperature
- Vehicle speed
- Fuel trim
- Calculated load
- Throttle position

Why freeze frame helpful:

- Shows operating conditions when fault occurred
- Helps identify patterns
- **Example:** P0300 misfire at idle vs at highway speed suggests different causes

Students practice:

- Access freeze frame data

- Record key parameters
- Understand significance

Pass Criteria:

- ✅ Can read fault codes successfully
- ✅ Can interpret code descriptions
- ✅ Understands code status meanings
- ✅ Can access freeze frame data
- ✅ Records diagnostic information properly

Hour 3 (3:00 PM - 4:00 PM): Live Data & Code Clearing

Activity 1: Viewing Live Data (30 min)

- **Access live data:**
  1. Select "Live Data" or "Data Stream"
  2. Scanner shows real-time sensor values
  3. May show all parameters or allow selection

Common live data parameters:

- **Engine RPM:** 850 RPM (idle), increases with throttle
- **Coolant Temp:** 85-95°C (normal operating)
- **Intake Air Temp:** Near ambient temperature
- **Throttle Position:** 0% idle, 100% full throttle
- **Engine Load:** 15-25% idle, higher under load
- **Short Term Fuel Trim:** -10% to +10% (normal)
- **Long Term Fuel Trim:** -10% to +10% (normal)
- **O2 Sensor B1S1:** Switching 0.1V to 0.9V (narrowband)
- **Vehicle Speed:** 0 km/h (stopped)
- **MAF/MAP:** Varies with throttle

Students observe:

- Engine at idle (live data)
- Have assistant rev engine gently
- Watch parameters change
- **Note relationships:**
  - Throttle opens → RPM rises
  - RPM rises → MAF increases

  - Engine warms → Coolant temp rises

Understanding normal vs abnormal:

- **Normal coolant temp:** 85-95°C
  - Much lower = thermostat stuck open (P0128 code)
  - Much higher = overheating
- **Normal O2 sensor:** Switches rapidly
  - Stuck high/low = sensor failure
- **Normal fuel trim:** ±10%
  - High positive = system running lean (adding fuel)
  - High negative = system running rich (reducing fuel)

NOTE: Detailed data interpretation = Diagnostics Track

- Fundamentals = observe and understand basics only

Activity 2: Clearing Codes (15 min)

- **When to clear codes:**
  - ✅ AFTER repair completed
  - ✅ To verify if code returns (after repair)
  - ❌ NOT to turn off light without fixing!
  - ❌ NOT before diagnosing (lose freeze frame data)

Clearing procedure:

1. Select "Erase Codes" or "Clear DTCs"
2. Scanner asks "Are you sure?" (usually)
3. Confirm
4. Scanner sends clear command
5. ECU erases codes
6. MIL turns off
7. Monitors reset to "Not Ready"

After clearing:

- Codes erased (no stored codes)
- MIL off
- Freeze frame erased
- Monitors not ready
- **If problem still exists:**
  - Code will return
  - May take one drive cycle or several

    - MIL will illuminate again

Students practice:

- Clear codes on practice vehicle
- Verify MIL off
- Check monitor status (all not ready)
- Understand consequences

Activity 3: Monitor Status Check (15 min)

- **Access monitor status:**
    - Select "Monitor Status" or "Readiness Monitors"
    - Shows list of monitors
    - Each shows "Ready" or "Not Ready"

Common monitors:

- Misfire Monitor
- Fuel System Monitor
- Comprehensive Component Monitor (always ready)
- Catalyst Monitor
- Oxygen Sensor Monitor
- EGR Monitor (if equipped)
- EVAP System Monitor
- Secondary Air (if equipped)

Drive cycle concept:

- **Purpose:** Complete all monitors
- **Requirements:** Varies by vehicle
- **Typically needs:**
    - Cold start
    - Idle period
    - Acceleration
    - Steady cruise
    - Deceleration
- **May take 50-100 km** to complete all monitors

Why it matters:

- Emission testing requires monitors ready
- After battery disconnect or codes cleared
- Must complete drive cycle before test

Pass Criteria:

- ✅ Can view live data parameters
- ✅ Understands normal parameter ranges
- ✅ Can clear codes properly
- ✅ Understands when to clear codes
- ✅ Can check monitor status
- ✅ Understands drive cycle concept

DAY 24 ASSESSMENT:

Practical Skills Checklist:

- [ ] Locates OBD-II connector
- [ ] Connects scanner correctly
- [ ] Achieves communication with vehicle
- [ ] Navigates scanner menus
- [ ] Reads fault codes
- [ ] Interprets code descriptions
- [ ] Views freeze frame data
- [ ] Views live data parameters
- [ ] Clears codes properly
- [ ] Checks monitor status
- [ ] Disconnects scanner properly

Knowledge Check (Verbal):

- What does OBD-II stand for?
- What do the letters in a fault code mean (P, B, C, U)?
- Give example: What is P0301?
- When should you clear codes?
- What is freeze frame data?
- Why do monitors reset after clearing codes?